FRM4BIOMASS Toulouse meeting
WP2 - BIOMASS R package

A. Bailly, D. Lamonica, M. Réjou-Méchain

UMR AMAP, IRD

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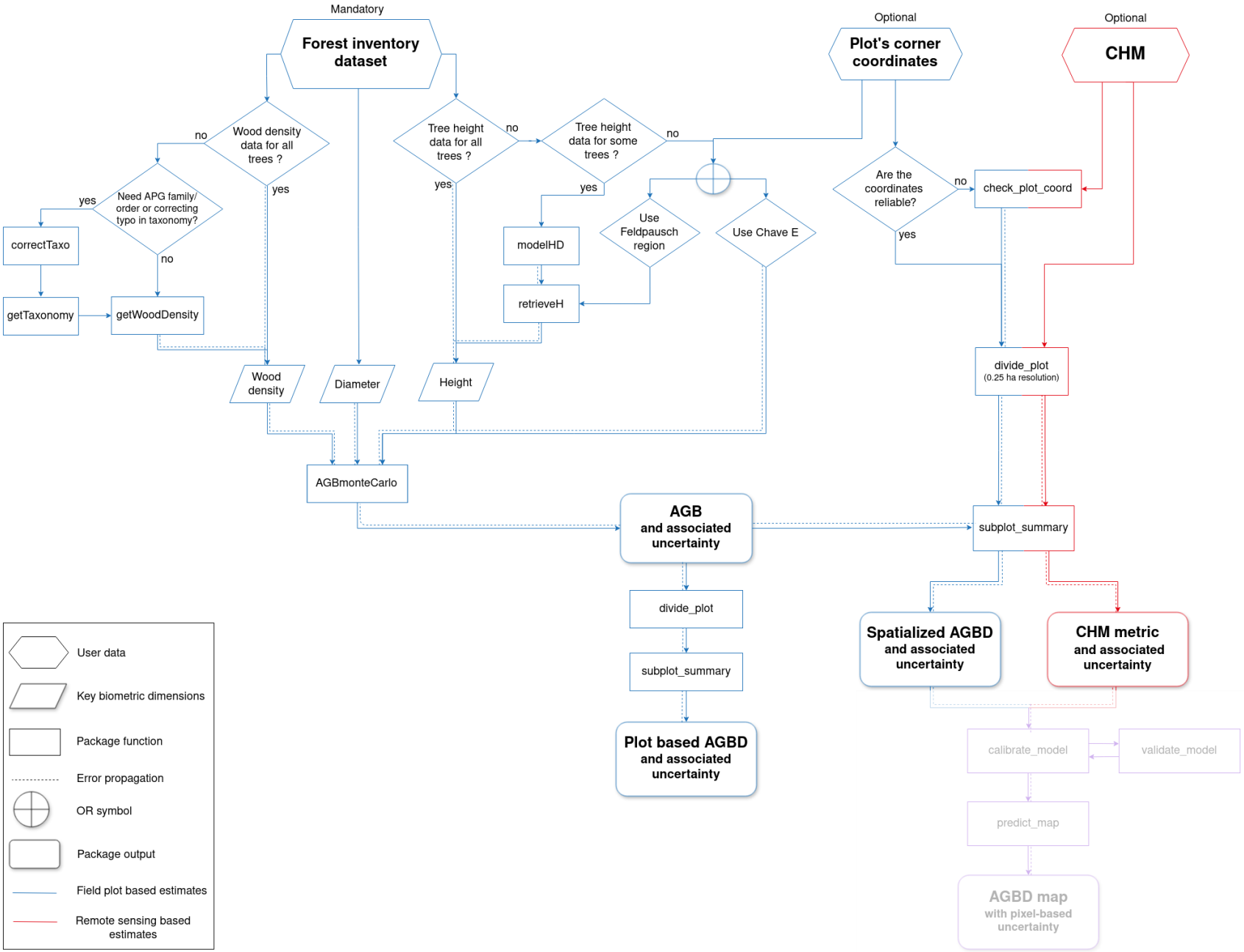
BIOMASS R package FRM4BIOMASS



FRM4BIOMASS: what has been done

Propagation of uncertainties in the H-D model

Since the version 2.2.6:

Building bayesian Height-Diameter models

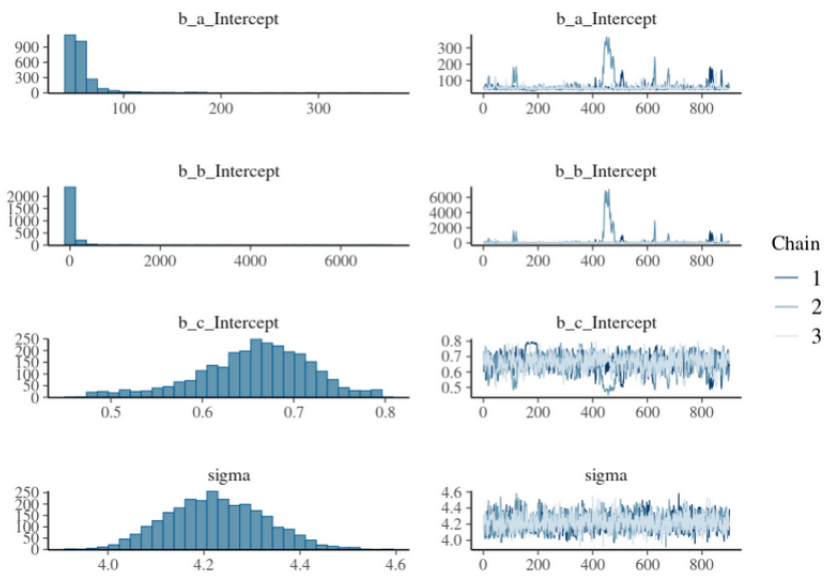
Since version 2.2.6, BIOMASS enables to build bayesian H-D models using the brms package.

As compiling the Stan programme may take some time the first time around, we recommend setting the 'useCache' argument to TRUE in the `modelHD()` function. This will save the model as a .rds file in the defined cache path, meaning that next time, the model will simply be loaded and updated, bypassing the compilation stage.

```
# First, define the user cache path
createCache("the_path_to_your_cache_folder")
# Or
options("BIOMASS.cache" = "the_path_to_your_cache_folder")

brm_model <- modelHD(
  D = NouraguesHD$D, H = NouraguesHD$H,
  method = "log2",
  bayesian = TRUE, useCache = TRUE)

H_brm_model <- retrieveH(
  D = NouraguesTrees$D,
  model = brm_model)
NouraguesTrees$H_model <-  H_brm_model$H
```

If the **Michaelis-Menten or Weibull methods are used**, you must pay close attention to the returned models as the algorithm is likely to identify numerous **local minima for the estimated parameters**. Here is an example of the Weibull method being used with the default arguments:

```
brm_model <- modelHD(
  D = NouraguesHD$D, H = NouraguesHD$H,
  method = "weibull",
  bayesian = TRUE, useCache = TRUE)

plot(brm_model$model)
```



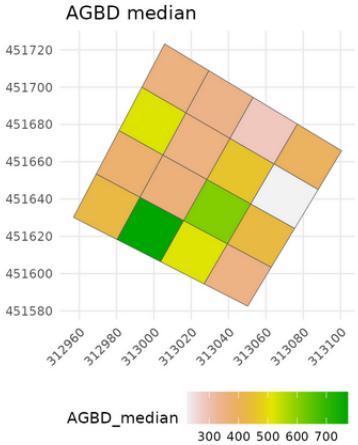
Propagation of uncertainties in spatialized metrics

Since version 2.2.7:

Summarising the AGBD and its uncertainties

If you have obtained the **AGB estimates and their uncertainty using the 'AGBmonteCarlo()'** function (see the Estimating stand biomass vignette, you can provide the output matrix of this function using the `AGB_simu` argument.
The function also handles uncertainty in corner coordinates if the `subplots` argument provided has been created using the `sd_coord` argument (optional):

```
subplot_AGBD <- subplot_summary(
  subplots = sd_coord_subplots,
  AGB_simu = error_prop$AGB_simu # error_prop has been created in the previous vi
)
#> As 'AGB_simu' contains 200 simulations, and 'subplots$simu_coord' contains 50
#> [[1]]
```

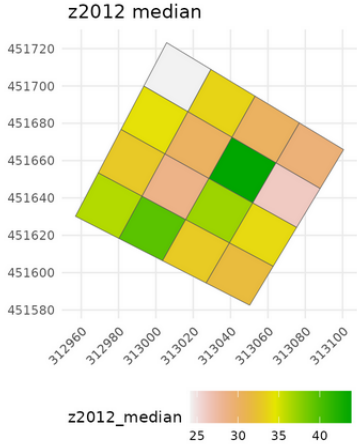


The simulated AGBs are therefore summarised in **AGBD** (i.e. AGB per hectare), and the function returns an additional output containing all the simulations performed: `$long_AGB_simu`.

Summarising LiDAR metrics

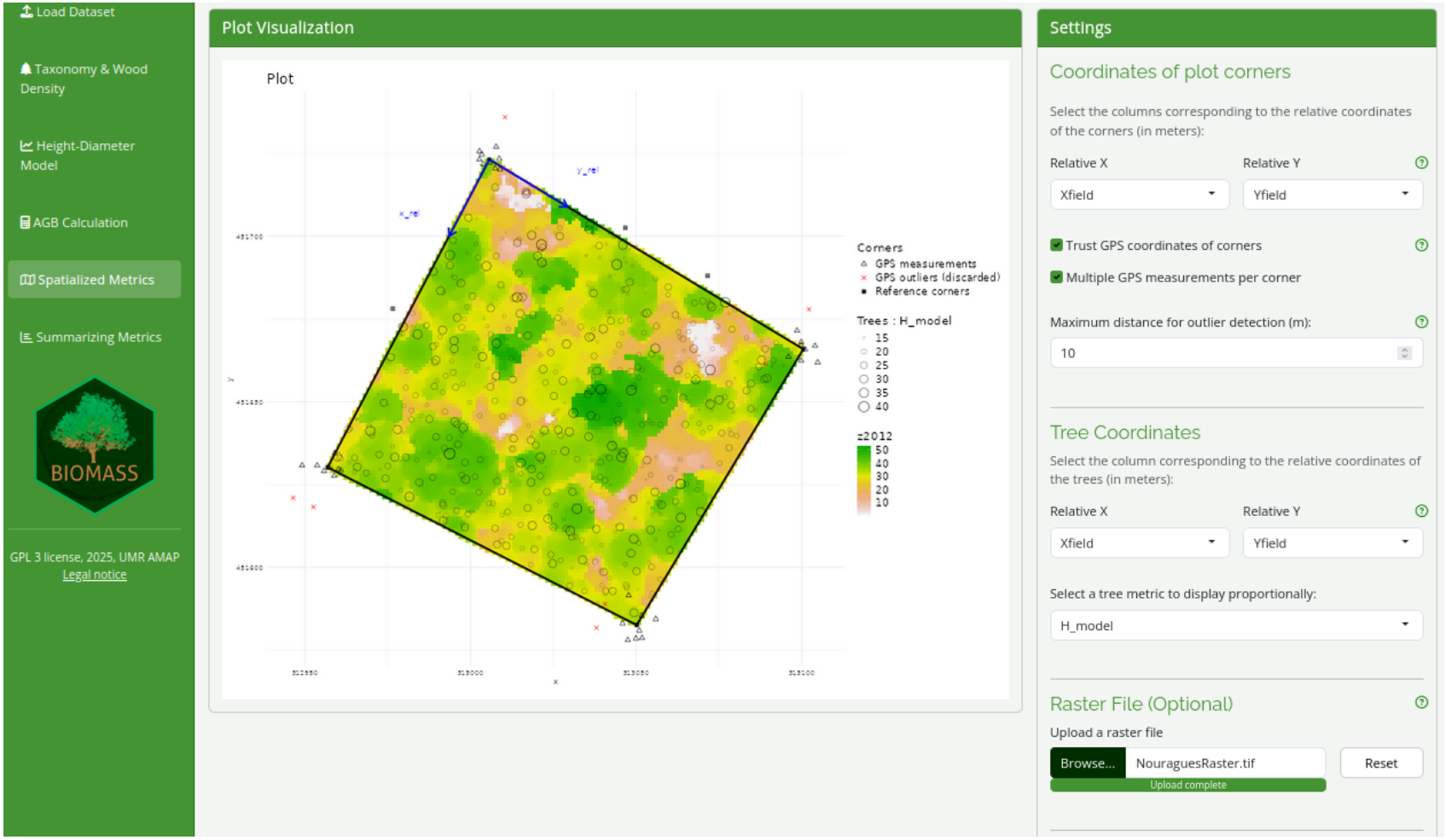
`subplot_summary()` can also **summarise the metric contained in a raster** (typically a CHM raster obtained from LiDAR data) by providing the `ref_raster` and the `raster_fun` arguments:

```
raster_summary <- subplot_summary(
  subplots = subplots,
  ref_raster = nouraguesRaster, raster_fun = median)
#> Extracting raster metric...Extracting raster metric done.[[1]]
```

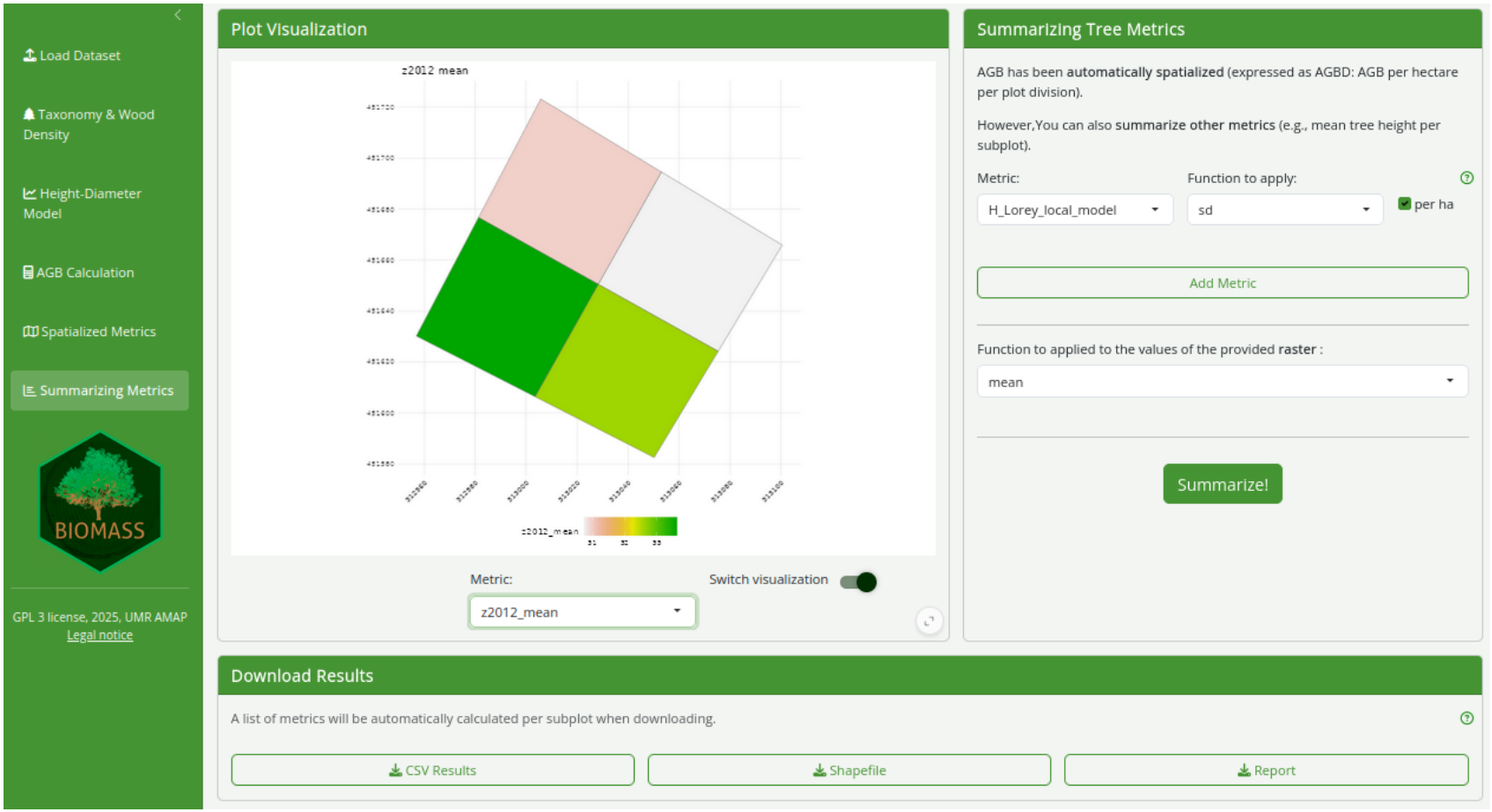


Shiny app for BIOMASS

The BIOMASS app now enables users to obtain spatialised estimates of AGBD and LiDAR metrics, as well as any other tree metrics.



Shiny app for BIOMASS

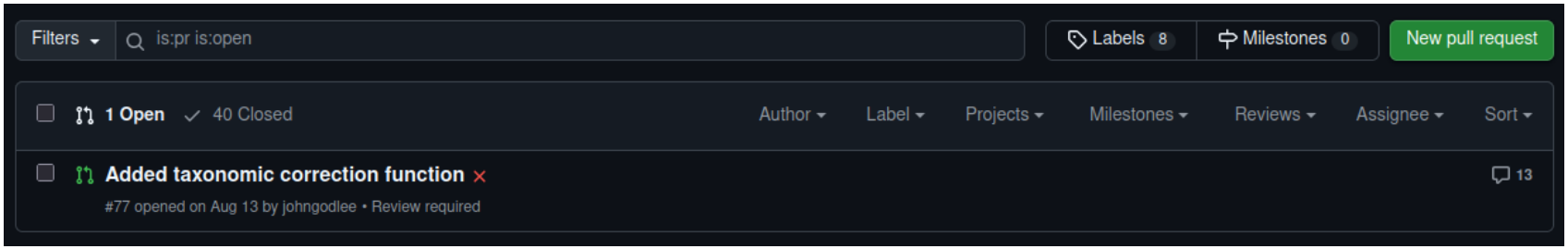


Propagation of plot corner uncertainties is not allowed in BIOMASSapp.

FRM4BIOMASS: ongoing work

updating taxonomy correction

Contribution by John Goodlee using the World Flora API.



Good work, but still some issues to preserve backward compatibility

updating the Global Wood Density Database v.2

Beyond species means – the intraspecific contribution to global wood density variation

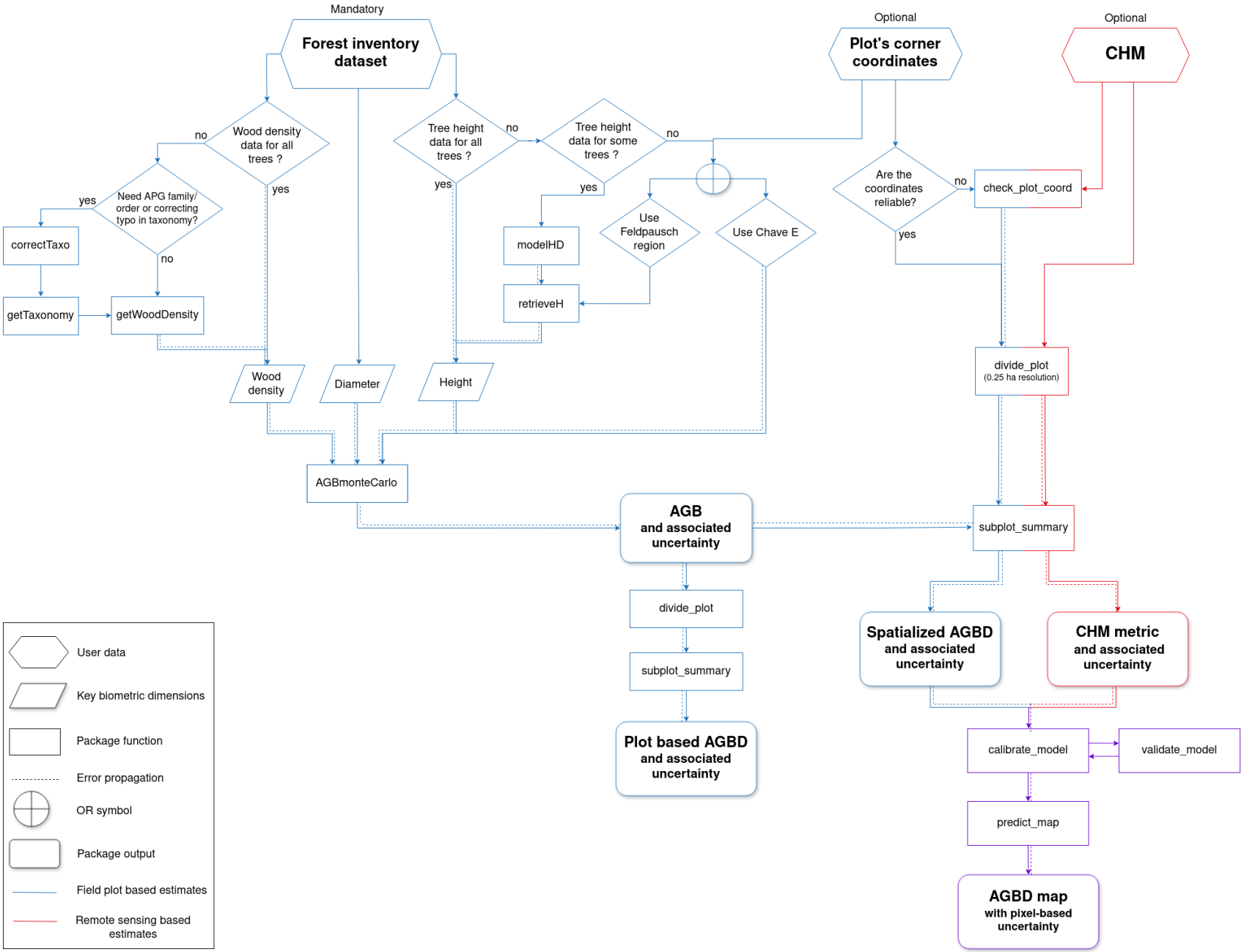
Fabian Jörg Fischer[1 2 3], Jérôme Chave[2], Amy Zanne[4], Tommaso Jucker[1], Alex Fajardo[5 6 7], Adeline Fayolle[8 9], Renato Augusto Ferreira de Lima[10], Ghislain Vieilledent[11 12 13], Hans

Based on the GWDD v.2, Fabian used bayesian hierarchical models to provide wood density estimates and the associated uncertainties at the species- genus- and family-level.

family	genus	species	wsg	sd	level_tax
Fabaceae	Abarema	Abarema cochliocarpos	0.7595	0.0947	species
Fabaceae	Abarema	Abarema jupunba var. jupunba	0.5971	0.0722	species
Fabaceae	Abarema	NA	0.6743	0.1036	genus
Fabaceae	NA	NA	0.6783	0.1216	family

Code in review, will be implemented in BIOMASS very soon.

CHM-AGBD model calibration (V3)



CHM-AGBD model calibration: proposed statistical framework

- geostatistical model with SPV-I/C (SPatially Varying Intercept/Coefficients) to integrate spatial correlation:
- $y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$ with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$
- references

Remote Sensing of Environment 169 (2015) 113–127

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

LiDAR based prediction of forest biomass using hierarchical models with spatially varying coefficients

Chad Babcock [a], Andrew O. Finley [b,*], John B. Bradford [c], Randall Kolka [d], Richard Birdsey [e], Michael G. Ryan [f]

Remote Sensing of Environment 318 (2025) 114557

Contents lists available at ScienceDirect

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journal homepage: www.elsevier.com/locate/rse

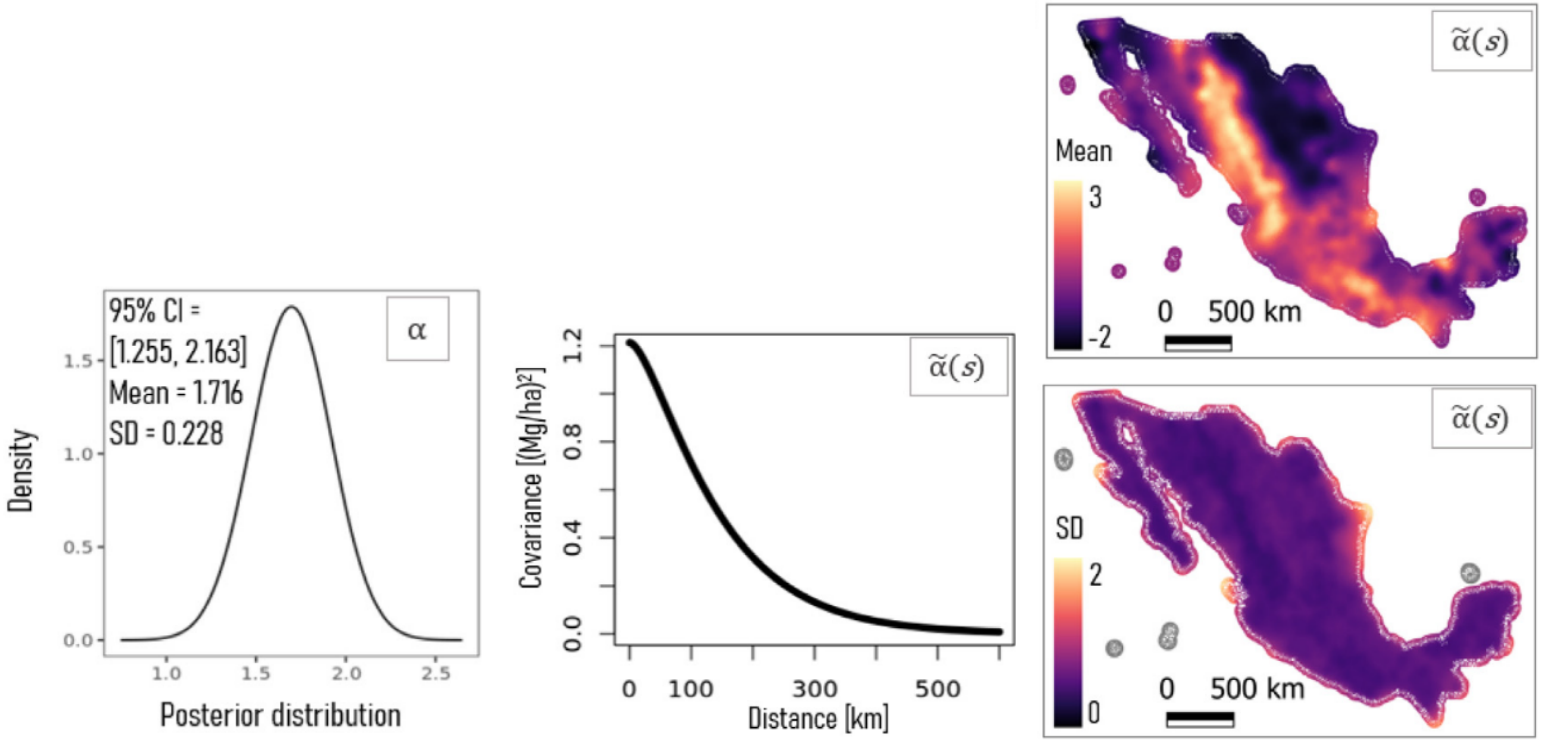
A geostatistical approach to enhancing national forest biomass assessments with Earth Observation to aid climate policy needs

Neha Hunka [a,*], Paul May [b], Chad Babcock [c], José Armando Alanís de la Rosa [d], Maria de los Ángeles Soriano-Luna [d], Rafael Mayorga Saucedo [d], John Armston [a], Maurizio Santoro [e], Daniela Requena Suarez [f], Martin Herold [f], Natalia Málaga [f], Sean P. Healey [g], Robert E. Kennedy [h], Andrew T. Hudak [i], Laura Duncanson [a]

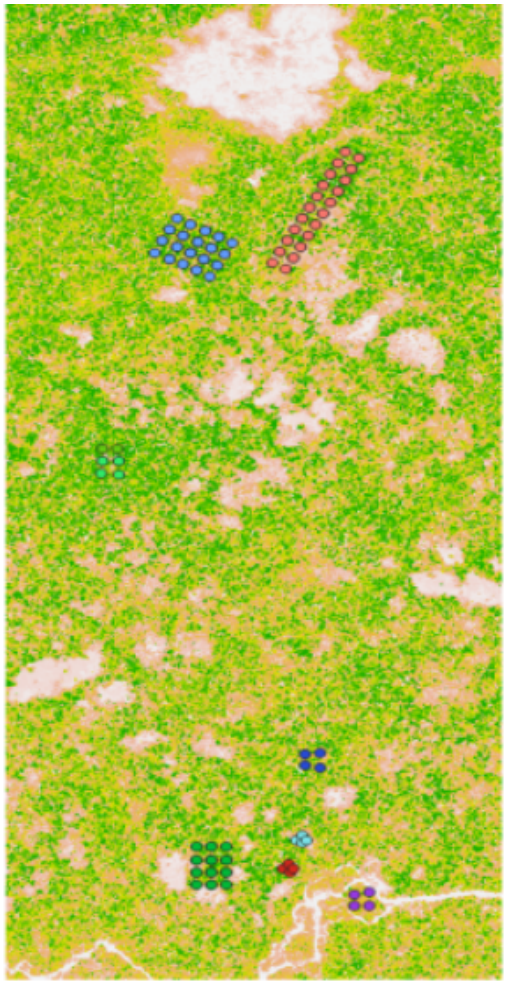
CHM-AGBD model calibration: proposed statistical framework

$$y(s) = (\alpha + \tilde{\alpha}(s)) + (\beta + \tilde{\beta}(s)) \times x(s) + \epsilon(s)$$

with $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$

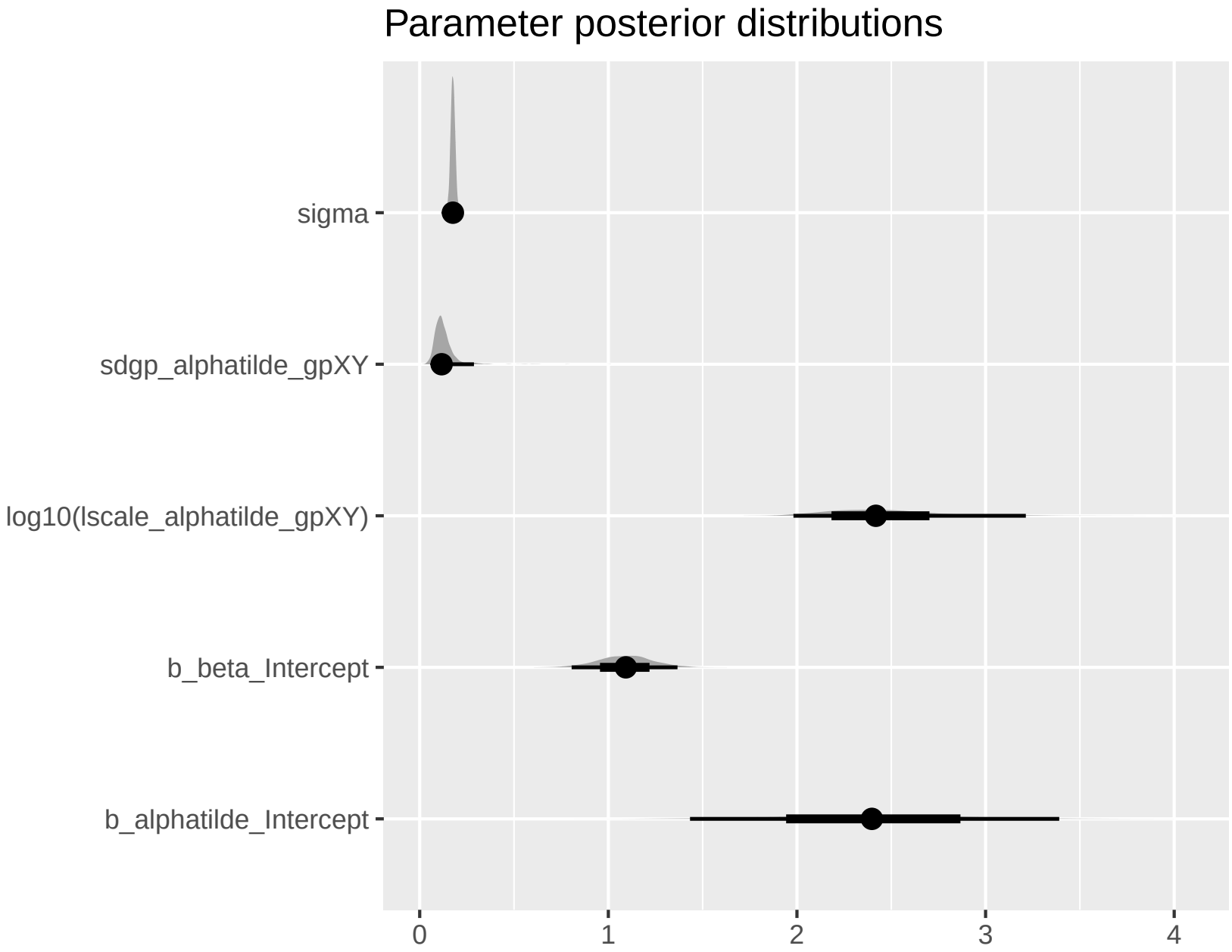


CHM-AGBD model calibration: example with Nouragues data

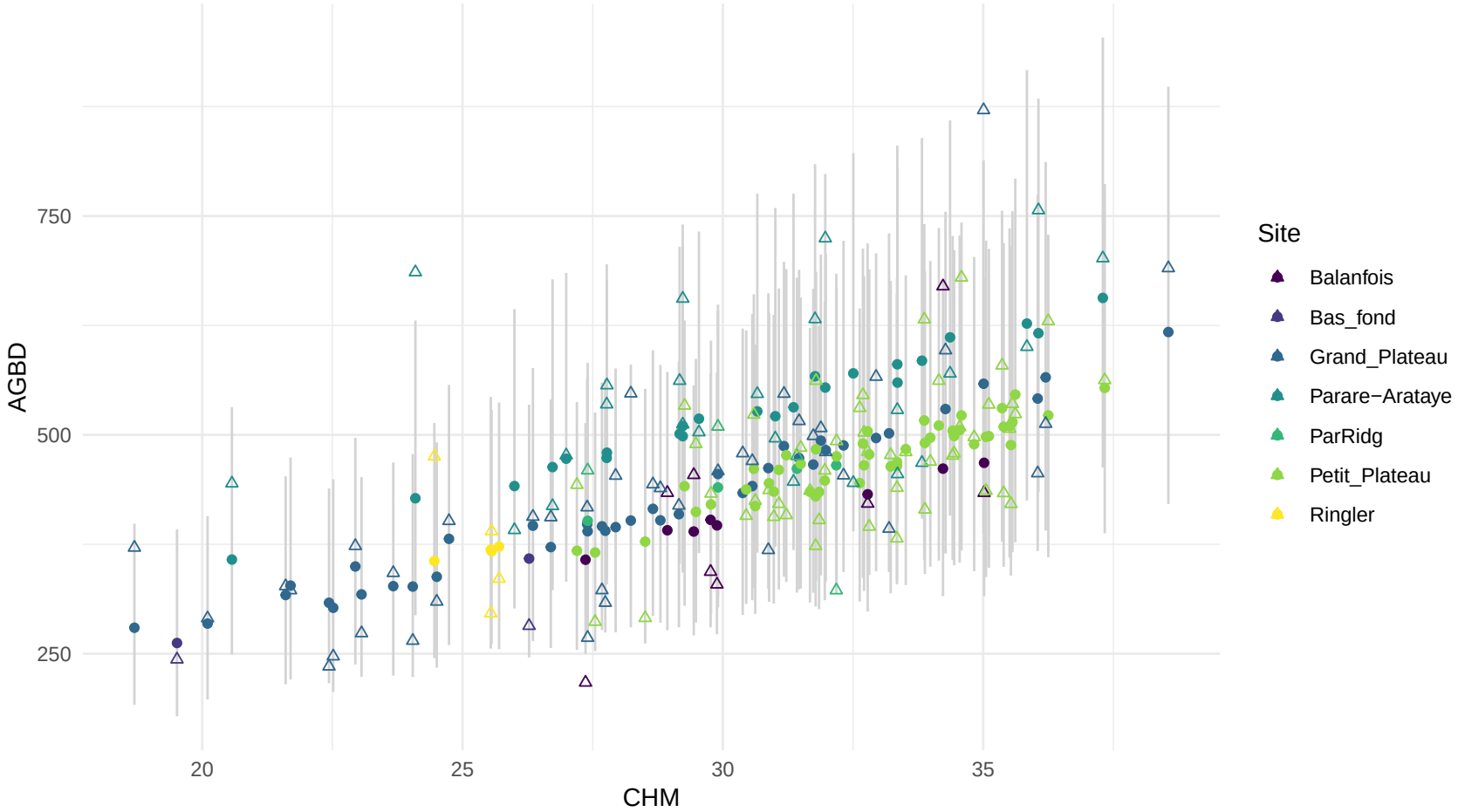


- SPV-I model
  $log(AGBD(s)) =$
  $(\alpha + \tilde{\alpha}(s)) + \beta \times log(CHM(s)) + \epsilon(s)$
  with
  $\tilde{\alpha}(s_1), ..., \tilde{\alpha}(s_n) \sim MVN(0, C_\alpha(s_i, s_j))$

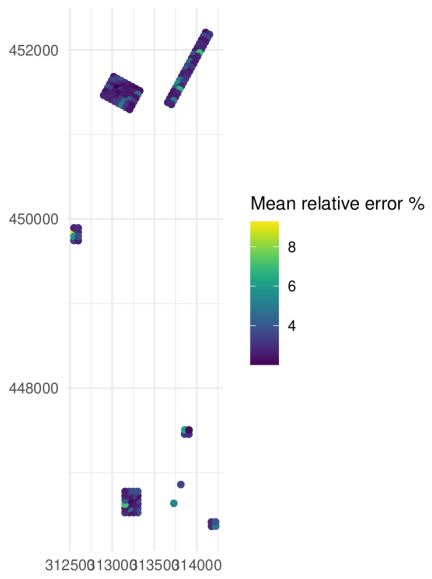
CHM-AGBD model estimates and plot prediction (1/2)

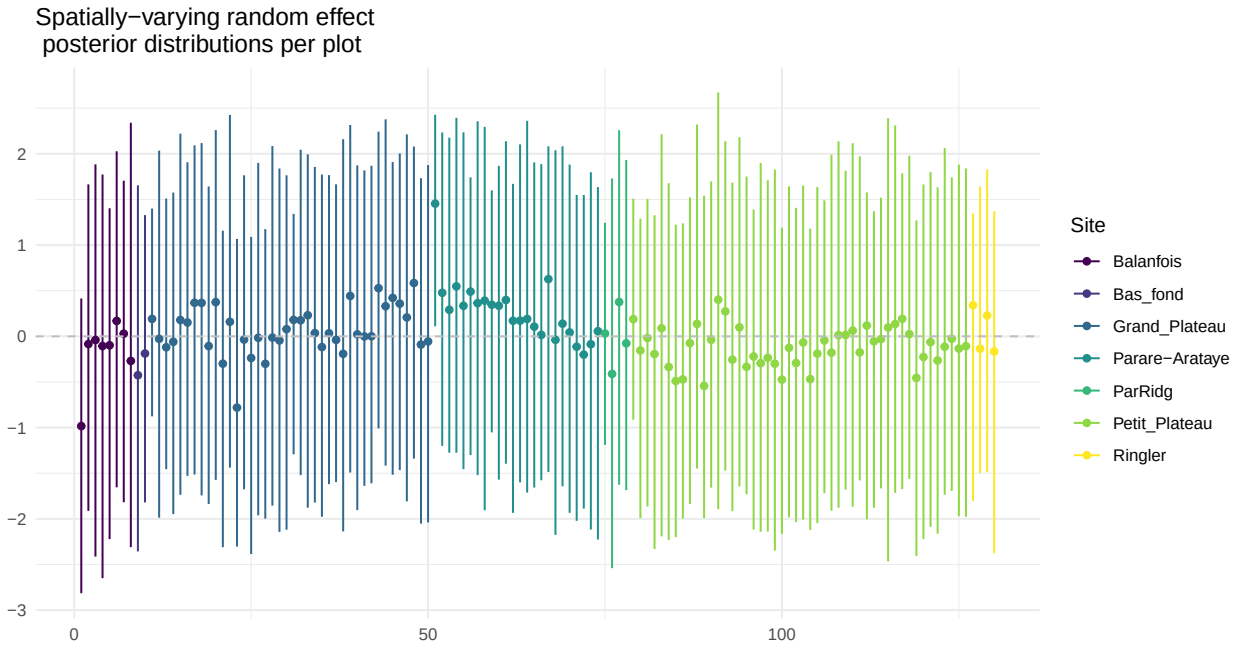


CHM-AGBD model estimates and plot prediction (2/2)

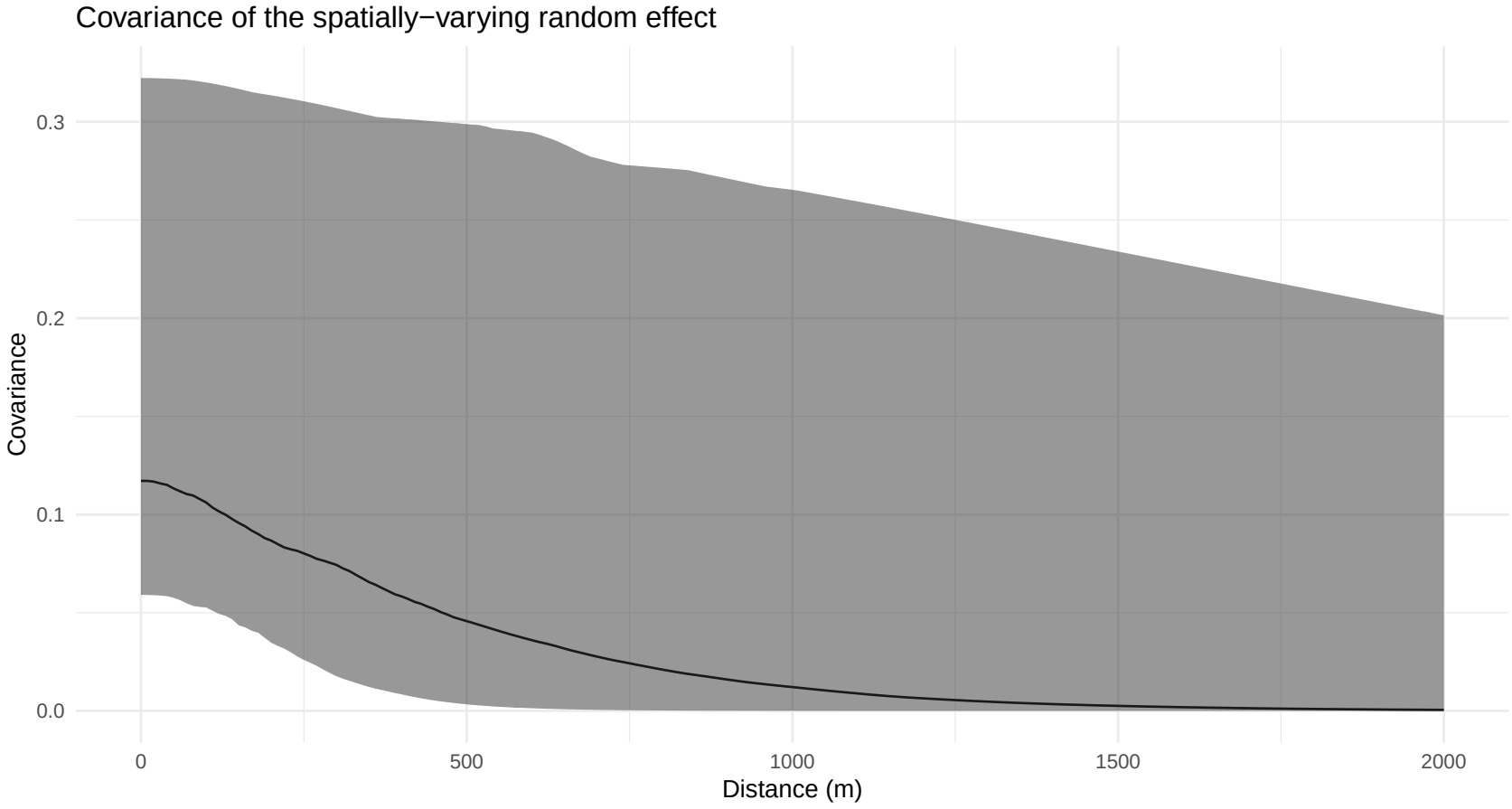


CHM-AGBD model errors: plots

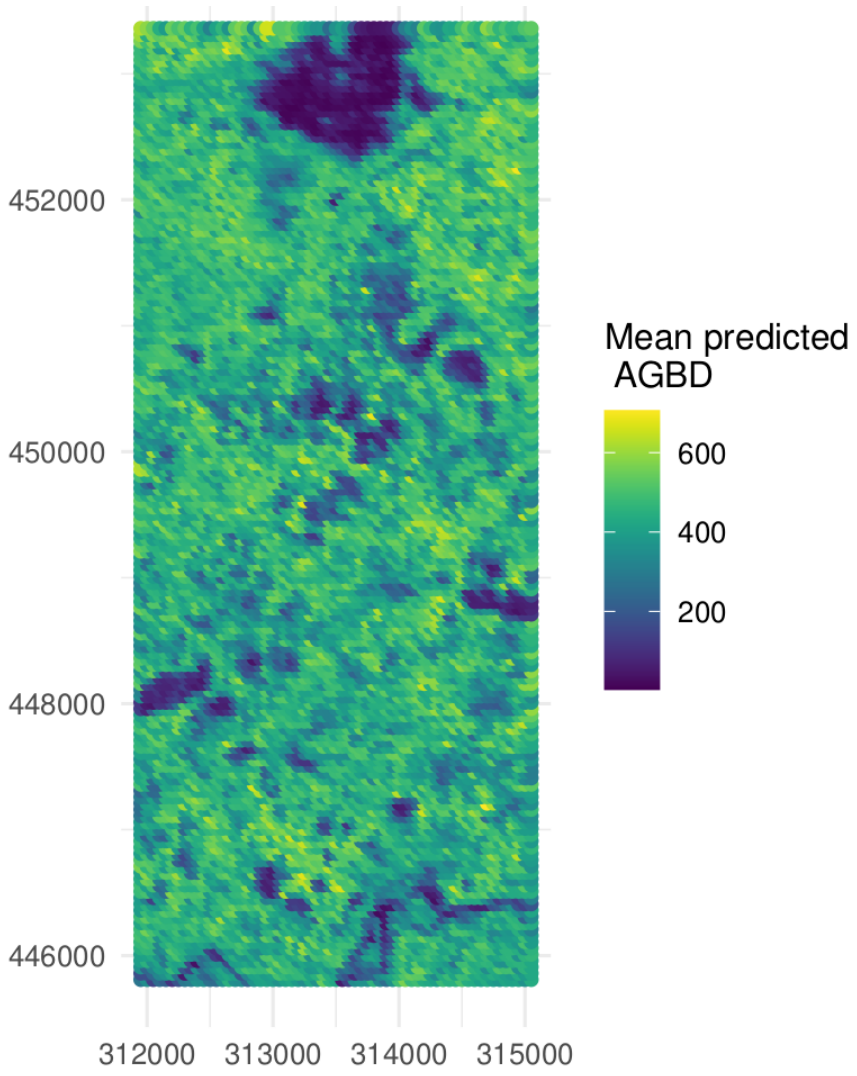


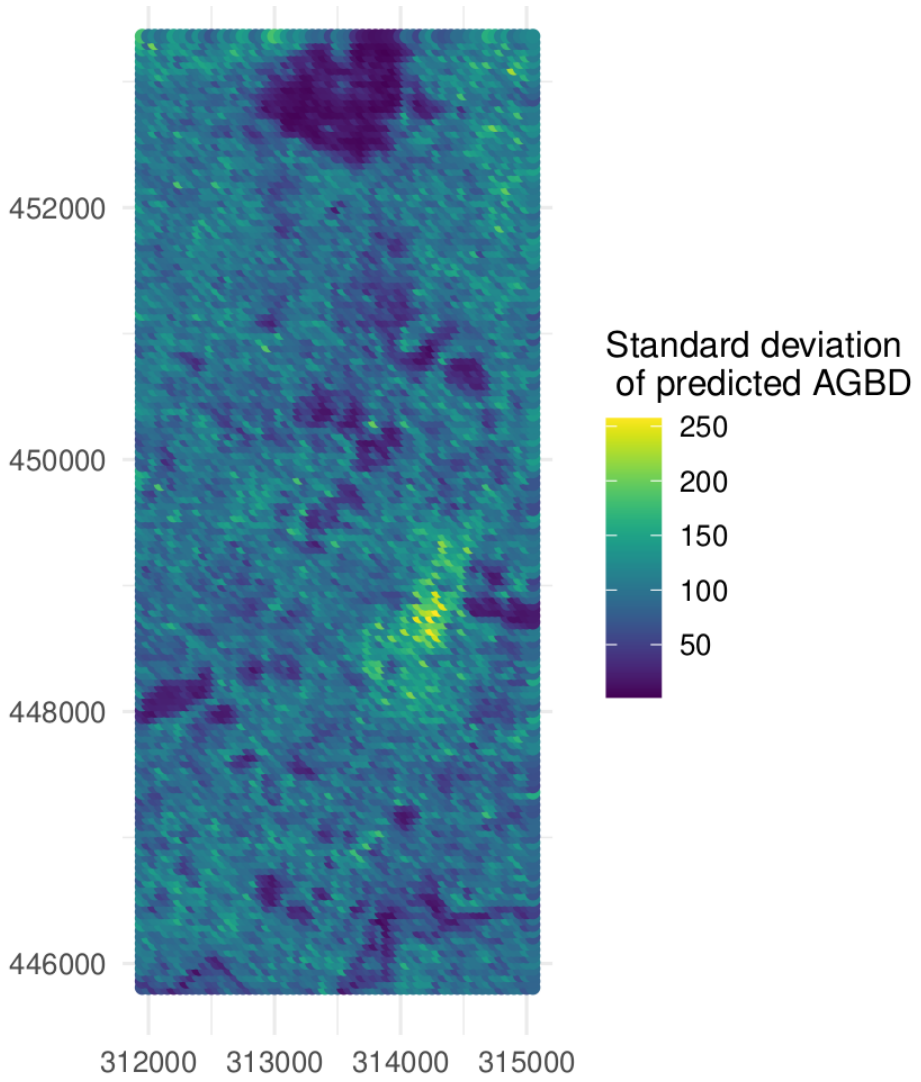


CHM-AGBD model landscape predictions (1/2)



CHM-AGBD model landscape predictions (2/2)



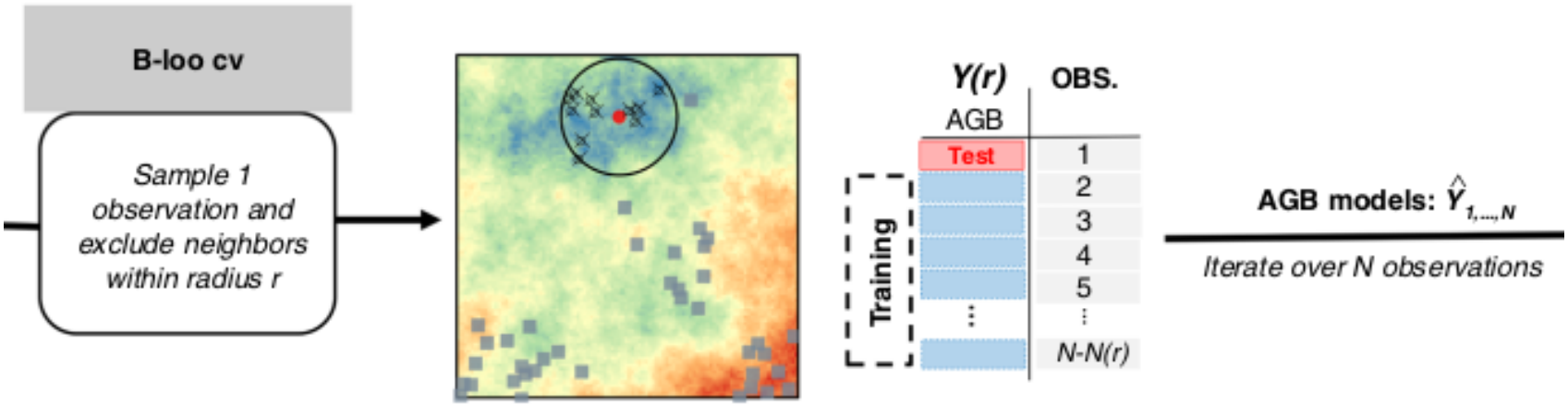


CHM-AGBD model calibration: implementation possibilities & difficulties

- brms package, STAN, geostat module in JAGS
- how to propagate AGBD estimates uncertainties, computation wise (eg Monte Carlo procedure, or directly into the model ?)
- pixel error propagation: Monte Carlo procedure
- future statistical development to use all the CHM spatial structure: better use of available information for a more robust & precise full spatial AGBD prediction (for a next major version)

CHM-AGBD model validation

- initially, proposed framework: spatial Leave-One-Out, Ploton et al. 2020 Nature Com.



- but computationally super intensive, so external validation (using independent dataset or splitting dataset) to be considered
- needs further discussion!

Final product: uncertainty sources & how to deal with them

$$U_{ref} = U_{Inst} + U_{Model} + U_{Location} + U_{Area} + U_{Representativeness}$$

- wood density, height, diameter U_{Inst}
- plot based AGB prediction: allometric relationship with Monte Carlo procedure U_{Model}
- plot based AGBD-CHM calibration: spatial structure with SPVI/C (Bayesian framework) U_{Model}
- full spatial AGBD prediction: plot based AGBD uncertainties with Monte Carlo procedure ? U_{Model}
- plot based AGB density & CHM computation: pixel error with Monte Carlo procedure $U_{Location}$ & U_{Area}

Perspectives

Short term perspectives - with Arthur

- new allometric relationship to predict AGB
- companion paper for V3 BIOMASS R package
- waiting for wood density database update
- update taxonomy correction, currently we do not deal with synonymy $\rightarrow$ waiting for Renato's package to be on CRAN
- error detection: outliers (diameter, height, wood density)

Long term perspectives - with ?

Temporal BIOMASS

- propagating joint errors on differentes dates, for plots and LiDAR
- technically challenging: package implementation and structure to integrate temporal dynamics
- approaches for allometric relationships and differences in AGBD, CHM ?

Thank you for your attention